

FastMCP Questions

1. FastMCP supports three transports: stdio, http, and sse. Read the deployment docs and explain when you'd pick stdio vs http, and what actually changes in your `mcp.run()` call to switch between them.

- **STDIO Transport:** is use when the MCP server is running locally and is used by a local client such as cloud desktop. `mcp.run()` automatically uses stdio.

Ex: A personal MCP server that allows Claude to access files from own computer.

- **HTTP Transport:** HTTP transport connects to MCP servers running as web services.

Ex:

```
HTTP
mcp.run(transport="http", host="0.0.0.0", port=8000)
```

HTTP allows network-based clients to connect and can support multiple clients.

In short:

stdio → local communication,

HTTP → network-based communication

2. FastMCP automatically turns some tool return values into “structured output” and others into plain text. Read the tools docs and explain the difference between what happens when a tool returns a plain string vs when it returns a dict or a Pydantic model, and why that distinction would matter to a client calling your tool.

FastMCP handles tool results differently depending on what the tool returns.

1. **Plain string:** When the tool returns a string, the result is normal text (plain text). The client mainly receives it as human readable text.

Example: “The email was sent successfully”

2. **Dict / Pydantic model:** if tool return output as a structured output, where the information is saved in different field.

Example:

```
return {"name": "Datta",
        "age": 21}
```

Instead of receiving one sentence, gets separate values like:

name → Datta

age → 21